



Introduction to Statistics

Statistics is a branch of mathematics that helps us collect, organize, analyze, and interpret data to make informed decisions. In **data analytics**, statistics provides the tools and techniques to extract insights and patterns from data.

Why is Statistics Important in Data Analytics?

- Helps in **understanding data**
- Identifies **trends, patterns, and anomalies**
- Supports **decision-making** with evidence

There are two type :

1. Descriptive statistics
2. Inferential statistics

Which type of statistics should be used for data analysis?

Descriptive Statistics

Provides a clear overview of data characteristics through measures like mean and standard deviation.



Inferential Statistics

Makes predictions about a population based on sample data using hypothesis testing.

Descriptive Statistics

Used to **summarize and describe** the features of a dataset.

- **Key Concepts:**
 - **Measure of Central Tendency[Mean, Median, Mode]:** The mean is the average, the median is the middle, and the mode is the most common.
 - **Measure of dispersion [Variance, Standard Deviation]:** These tell us how spread out or close together our data points are.
 - **Data Distribution :** Describes the **shape and spread** of data.

Measure of Central Tendency

Imagine you have the following test scores of 7 students:

Scores = [65, 70, 75, 80, 85, 85, 90]

Mean (Average)

$$\begin{aligned}\text{Mean} &= 65 + 70 + 75 + 80 + 85 + 85 + 90 / 7 \\ &= 550 / 7 \approx 78.57\end{aligned}$$

Median (Middle Value)

When values are in order, the **median** is the middle one.

Scores [sorted]=[65,70,75,80,85,85,90]

The middle value [4th position] is: **80**

Mode (Most Frequent Value)

The value that occurs most often.

85 appears twice, others only once\text{85 appears twice, others only once} 85

appears twice, others only once.

◊ **Mode = 85**

Measure of dispersion [Variance, Standard Deviation]

1. Variance

Average of the **squared differences** from the mean.

Population :

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

10,11,12,13,15 mean=12.5

[(10-12.5)**2 + (11-12.5)**2 +]/5

Why use it?

- Quantifies **total variation** in the data.
- Sensitive to outliers (because of squaring).

2. Standard Deviation

Square root of variance.

$$\sigma = \sqrt{\sigma^2} \quad \text{or} \quad s = \sqrt{s^2}$$

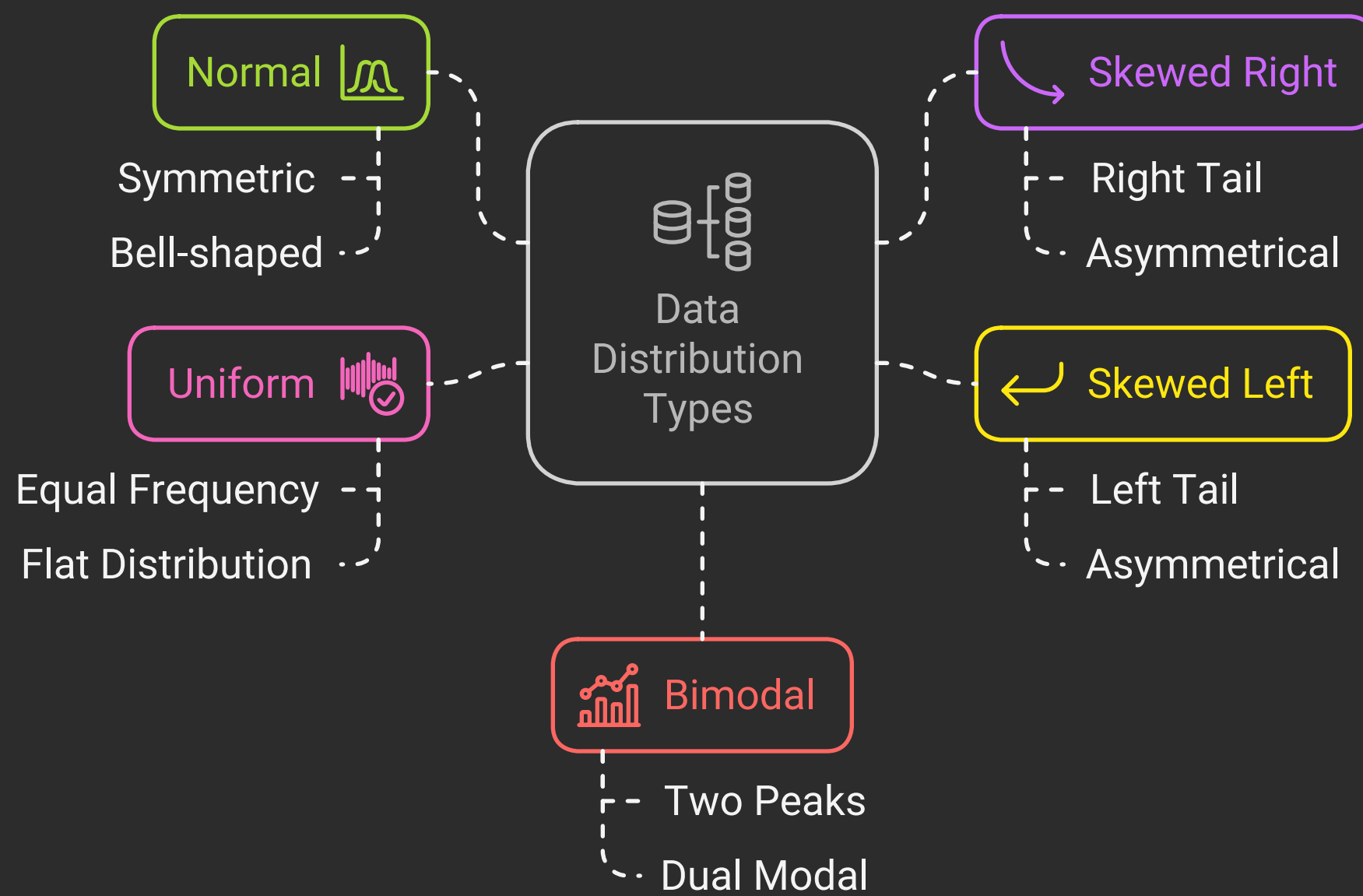
Why use it?

- Measures **average distance from the mean**.
- Same unit as the original data, making it easier to interpret.
- Lower SD = data tightly clustered around the mean; Higher SD = data is spread out.

Data Distribution

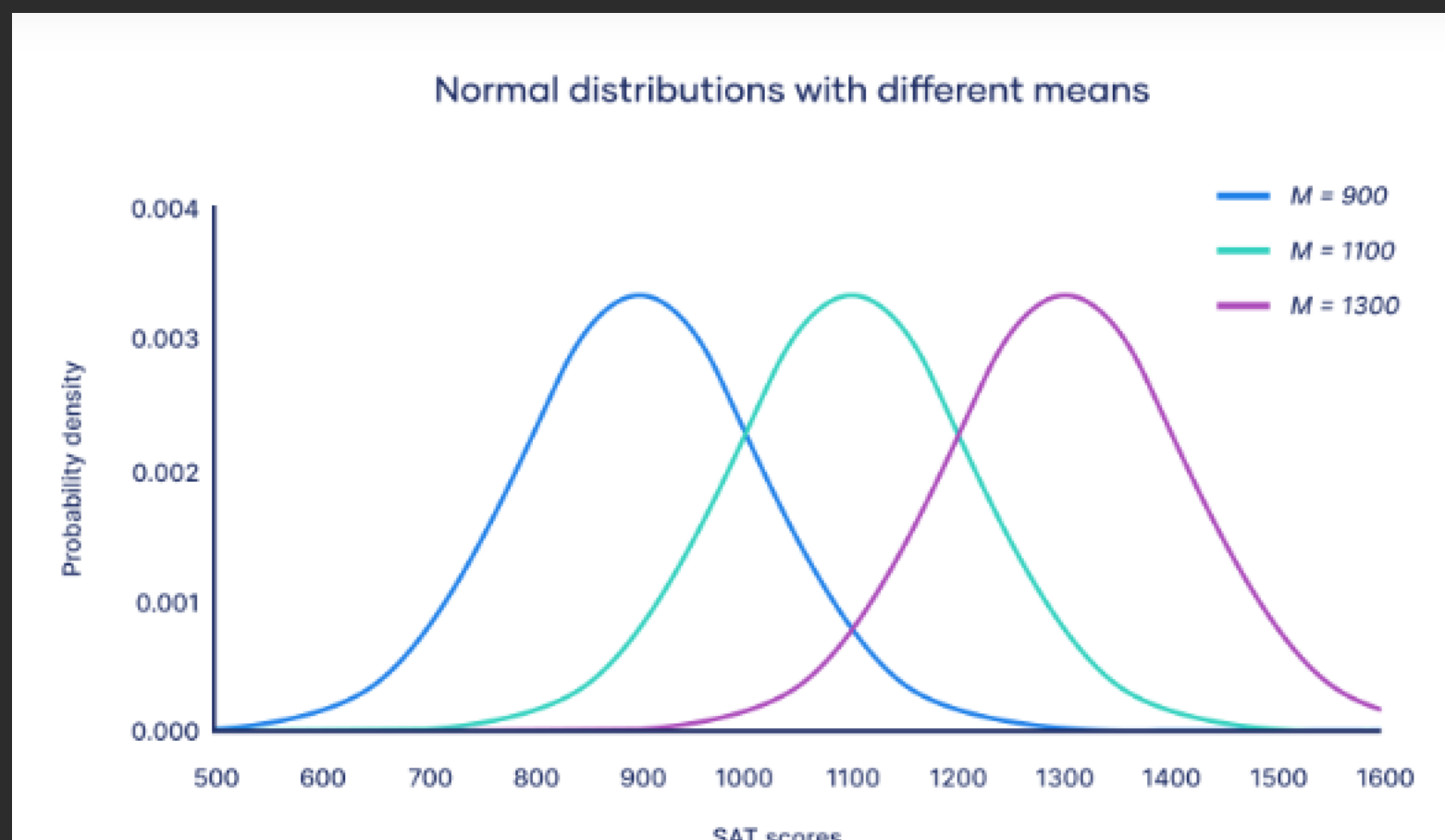
Describes the **shape and spread** of data.

Understanding Data Distribution Types



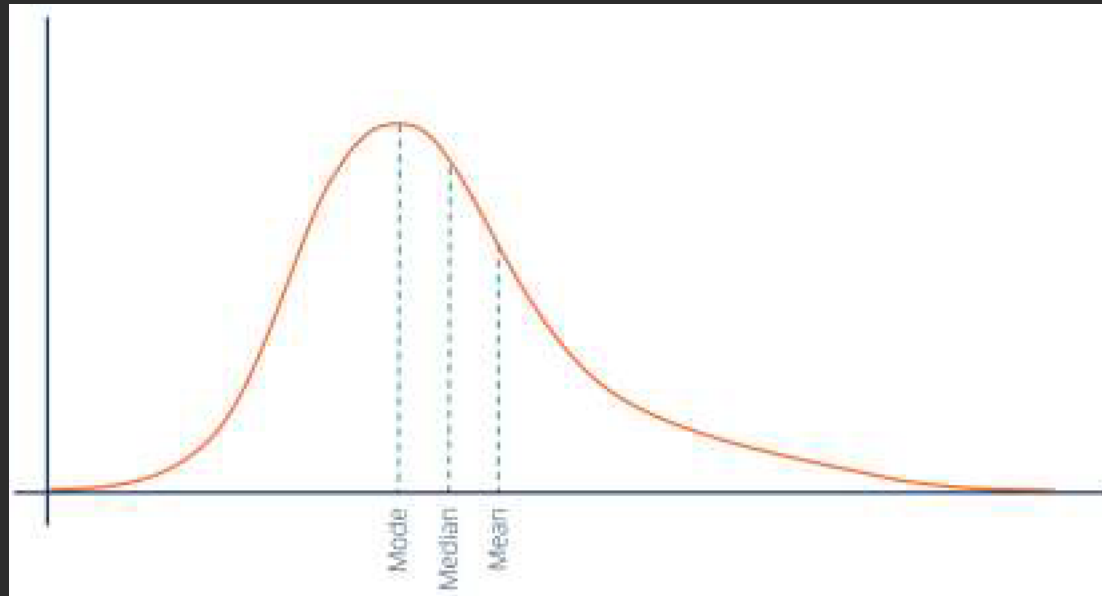
Normal Distribution

- **Shape:** Bell-shaped, symmetric
- **Mean = Median = Mode**
- Most data points are near the center; fewer at the extremes.
- Example: Heights, blood pressure



Right-Skewed Distribution (Skewed Right)

- **Shape:** Tail stretches to the right
- **Mean > Median > Mode**
- Most data values are on the **left**, but a few high values pull the mean right.
- Skewness > 0.
- **Example:** Income distribution

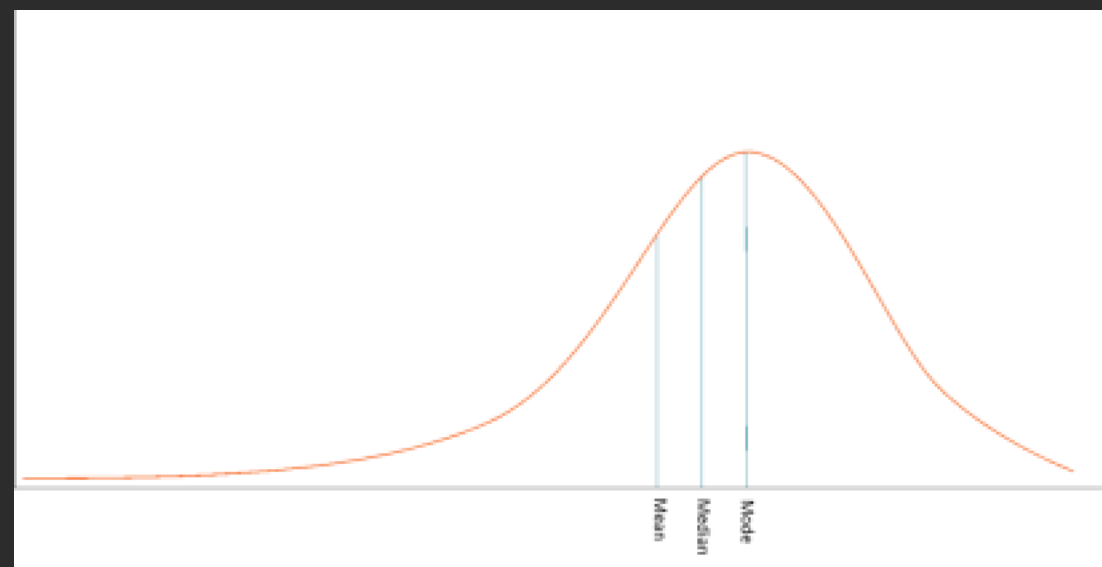


Skewness

Skewness measures the asymmetry or deviation from symmetry in the distribution of data.
Skewness = 0. -Normal distribution.

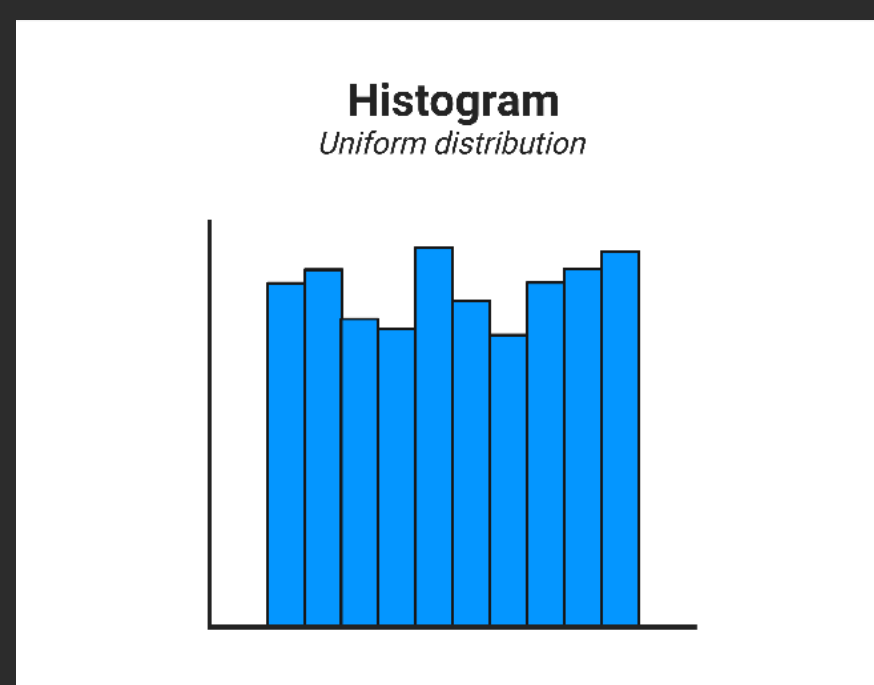
🔙!! Left-Skewed Distribution (Skewed Left)

- **Shape:** Tail stretches to the left
- **Mean < Median < Mode**
- Most data values are on the **right**, but a few low values pull the mean left.
- Skewness < 0.
- **Example:** Retirement age



📊 Uniform Distribution

- **Shape:** Flat; all values occur with equal frequency
- No obvious peak
- Every value has roughly the same chance
- **Example:** Rolling a fair die



⚠️!! Bimodal Distribution

- **Shape:** Two distinct peaks or "modes"
- Suggests data comes from two different groups
- **Example:** Exam scores from two different classes

